Pratik Nanaso Raut

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PROFESSIONAL SUMMARY

MCA candidate with strong foundation in software development principles and methodologies, Java, and C++ — a proactive learner eager to contribute to software engineering.

TECHNICAL SKILLS

Languages	Java, C, C++, Python (basics), JavaScript, SQL
Backend	Spring Boot, Spring Security, Hibernate, JDBC, REST APIs, JWT
Databases	MySQL, MongoDB
Tools	Git, GitHub, Maven, Postman, IntelliJ IDEA
Concepts	SDLC, Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Design Patterns, Problem Solving, Time Complexity Analysis, REST API Design, Database Optimization, RBAC, JWT Authentication, Microservice Fundamentals, API Security, Multi-Tenant Architecture, Multithreading
Exposure	AWS Fundamentals, AI/ML Fundamentals, Big Data Concepts

PROFESSIONAL EXPERIENCE

Software Developer Intern *Feb 2026 -- Ongoing*
SetTribe IT Solutions | Pune

- Developed and enhanced ERP backend modules using Java, Spring Boot, Hibernate, and REST APIs for employee and operational workflows.
- Integrated PostgreSQL database operations, validation logic, and API testing workflows using Postman.
- Collaborated in Agile sprint cycles, performing Git-based code reviews, bug fixes, feature implementation, and deployment support.

ACADEMIC PROJECTS

Concurrent FTP Server -- Multi-Client File Transfer Protocol Server *C, POSIX Threads, TCP Sockets, Multithreading*

- Engineered concurrent client handling using POSIX threads over TCP sockets, supporting simultaneous file transfers without data corruption.
- Implemented thread-safe resource management using mutex and semaphore primitives, eliminating race conditions under concurrent load.
- Designed application-layer protocol over raw sockets handling authentication, file listing, upload, and download command sequences.

Customised Virtual File System (CVFS) -- In-Memory Linux-like File Manager *C, C++, Data Structures, OS Concepts*

- Simulated Linux file system operations (open, read, write, lseek) in C++ using inode-based metadata management and a custom shell interpreter.
- Achieved O(1) file lookup by designing hash table-based inode indexing with chaining for collision resolution.
- Implemented in-memory file allocation table using C++ classes managing file permissions, ownership, and hierarchical directory traversal.

ProcMemInspector -- Linux Process Memory Layout Analyzer *C, Linux Internals, /proc Filesystem*

- Developed a /proc filesystem parser in C to extract and display virtual memory layout of live Linux processes including stack, heap, data, BSS, and mapped library segments.
- Implemented process statistics monitoring by reading /proc/[pid]/maps and /proc/[pid]/status with structured output formatting.
- Analyzed virtual address space segmentation demonstrating memory isolation boundaries and shared library loading behavior.

TECHNICAL STRENGTHS

- Strong foundation in Data Structures, Algorithms, and Object-Oriented Design.
- Hands-on experience building concurrent, multi-tenant, and performance-optimized Java applications.
- Analytical problem-solving with ability to design and develop solutions under time-bound delivery constraints.
- Ability to rapidly learn unfamiliar technologies and deliver production-ready solutions under tight deadlines.

EDUCATION

Master of Computer Applications (MCA) *2026 | Pursuing*
Dnyansagar Institute of Management and Research, Pune (SPPU) 7.24 CGPA

Bachelor of Computer Applications (BCA) *2024*
C.T. Bora College, Shirur (SPPU) 73.34%

Higher Secondary Certificate (HSC) *2021*
Maharashtra State Board 75%